Operating Systems Becoming a Power User.

1. The Multi-OS Administration & Security Audit

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- **Technical Stack:** Linux (ISH/Bash), Windows (PowerShell), TCP/IP, and Virtualisation
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- This audit focuses on the deployment and management of a secure multi user environment. By simulating a corporate Linux infrastructure, I have demonstrated the ability to configure core system components, manage network connectivity, and enforce security policies essential for protecting organisational data.

- I implemented User Access Control (UAC) by creating specific groups.This ensures that only authorised personnel can access sensitive financial directories, adhering to the principle of Least Privilege.

- **Core Competencies Demonstrated**

➢ **Identity Management:** Provisioning of users and security groups via CLI.

➢ **Access Control:** Implementation of the Principle of Least Privilege (PoLP) using Linux permissions.

➢ **Infrastructure Analysis:** Utilising, diagnostic tools (Ping, Trace-route, NSLookup) to ensure network integrity.

➢ **Automation & Scripting:** Efficient file system management using Bash commands.

- **System Administration & User Provisioning**

```
localhost:~# mkdir -p ~/Portfolio/Operating_System_Becom
ing_a_Power_User
localhost:~# pwd
/root
localhost:~# ls
Ching          Portfolio      index.html
Ching number1  bot.py
localhost:~# cd Portfolio
localhost:~/Portfolio# ls
Operating_System_Becoming_a_Power_User
Technical_Support_Fundamentals
The_Bits_and_Bytes_of_Networking
localhost:~/Portfolio# cd
localhost:~# addgroup finance_dept
localhost:~# adduser stanley_admin
Changing password for stanley_admin
New password:
Retype password:
passwd: password for stanley_admin changed by root
localhost:~#
```

- **<u>Administrative Provisioning & Directory Architecture</u>**

➢ In this phase of the audit. I utilised the Alpine Linux (ISH) terminal to establish a secure multi-user environment.The following administrative actions were performed to satisfy the core security requirements of the project.

- **Directory Hierarchy:** Used the mkdir -p command to initialise a structured portfolio system. This demonstrates the ability to manage complex file system and maintain organisational data integrity.

- **Identity Management:** Successfully provisioned a new security group (finance_dept) and an administrative user account (stanley_admin)

- **Security Enforcement:**By assigning a unique, encrypted password to the new user, I ensured that local authentication protocols were met, preventing unauthorised access to the system root.

- **<u>Permission Management and & Data Security</u>**

- In the final stage of the security audit, I focused on the principle of Least Privilege. After establishing the user and group hierarchy I implemented specific file level permission to ensure that sensitive company data is only accessible to authorised personnel. This involves changing the ownership of critical files and modifying read/write/execute bit-masks to prevent unauthorised data exfiltration or accidental modification by non administrative users.

```
localhost:~# touch /home/stanley_admin/salary_details.tx
t
localhost:~# chown :finance_dept /home/stanley_admin/sal
ary details.txt
chown: /home/stanley_admin/salary: No such file or direc
tory
chown: details.txt: No such file or directory
localhost:~# chown :finance_dept /home/stanley_admin/sal
ary_details.txt
localhost:~# chmod 640 /home/stanley_admin/salary_detail
s.txt
localhost:~#
```

- The terminal output above confirms the final execution of the security audit within the ISH (Alpine Linux) environment. Despite initial syntax errors during path specification, I successfully executed the **chown** and **chmod** commands secure the sensitive salary_details.txt file.

- **Ownership:** The file was successfully assigned to the finance_dept group.

- **Permission Logic:** By applying **chmod** 640**,** the file is now restricted: the owner has read/write access, the finance group has read only access, and all other system users are completely blocked.

- **<u>Environmental Note & Troubleshooting</u>**

Objective	Tools Used	Outcome
Directory Architecture	mkdir, pwd, ls	Organised multi-tier file system
Identity Management	addgroup, adduser	Provisioned secure administrative roles
Access Control	chmod, chown	Applied Principle of Least Privilege (PoLP)

Network integrity	ping, nslookup	Verified Layer 3 and Layer 7 connectivity

- During the infrastructure analysis phase certain kernel-level network commands (**ip addr, ifconfig**) were restricted due to the emulated nature of the ISH environment.To ensure audit completion, I utilised host-level diagnostics to verify network integrity, demonstrating adaptability in a constrained virtualised environment